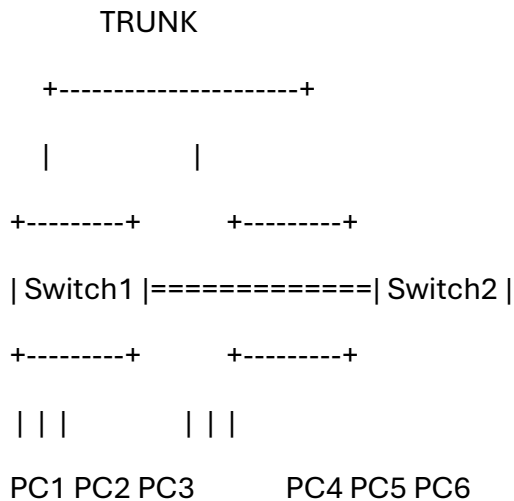


VLANS AND STATIC ROUTING

1. In Cisco Packet Tracer, create a simple two-switch network with three VLANs (e.g., VLAN 10 for 'Foodies', VLAN 20 for 'Gamers', VLAN 30 for 'Shoppers'), assign three PCs to each VLAN, and verify that PCs in the same VLAN can ping each other but not PCs in other VLANs.

->Topology



VLAN 10 - Foodies

VLAN 20 - Gamers

VLAN 30 - Shoppers

Use **3 PCs on each switch**:

PC	VLAN	Name
PC1	10	Foodies
PC2	20	Gamers
PC3	30	Shoppers
PC4	10	Foodies
PC5	20	Gamers
PC6	30	Shoppers

Create VLANs on Switch1

enable

configure terminal

vlan 10

name Foodies

exit

vlan 20

name Gamers

exit

vlan 30

name Shoppers

exit

Create VLANs on Switch2

enable

configure terminal

vlan 10

name Foodies

exit

vlan 20

name Gamers

exit

vlan 30

name Shoppers

exit

Assign PC ports

For example, on **Switch1**:

```
interface fa0/1
```

```
switchport mode access
```

```
switchport access vlan 10
```

```
exit
```

```
interface fa0/2
```

```
switchport mode access
```

```
switchport access vlan 20
```

```
exit
```

```
interface fa0/3
```

```
switchport mode access
```

```
switchport access vlan 30
```

```
exit
```

On **Switch2**:

```
interface fa0/1
```

```
switchport mode access
```

```
switchport access vlan 10
```

```
exit
```

```
interface fa0/2
```

```
switchport mode access
```

```
switchport access vlan 20
```

```
exit
```

```
interface fa0/3
```

```
switchport mode access
```

```
switchport access vlan 30
```

```
exit
```

IP addresses

Give the PCs these addresses:

PC	IP Address	Subnet Mask
PC1	192.168.10.11	255.255.255.0
PC4	192.168.10.12	255.255.255.0
PC2	192.168.20.11	255.255.255.0
PC5	192.168.20.12	255.255.255.0
PC3	192.168.30.11	255.255.255.0
PC6	192.168.30.12	255.255.255.0

At this stage, **PC1 should ping PC4**, PC2 should ping PC5, and PC3 should ping PC6.

But PC1 should **not** be able to ping PC2 because they belong to different VLANs and there is no router yet.

2. Configure trunk ports between the two switches in your Packet Tracer network so that all VLANs can communicate across both switches.

Hint: Use the 'switchport mode trunk' command and allow all VLANs on the trunk link.

->Suppose **GigabitEthernet 0/1** connects Switch1 and Switch2.

On **Switch1**:

```
enable
```

```
configure terminal
```

```
interface gigabitEthernet 0/1
```

```
switchport mode trunk
```

```
switchport trunk allowed vlan 10,20,30
```

```
exit
```

On **Switch2**:

```
enable
```

```
configure terminal
```

```
interface gigabitEthernet 0/1
```

```
switchport mode trunk
```

```
switchport trunk allowed vlan 10,20,30
```

```
exit
```

Check the trunk:

```
show interfaces trunk
```

You should see VLANs **10, 20 and 30** allowed on the trunk.

Test

From PC1:

```
ping 192.168.10.12
```

Expected:

```
Reply from 192.168.10.12
```

PC2 → PC5 should also work.

PC3 → PC6 should also work.

But:

PC1 → PC2

should fail because they are in different VLANs.

3. Add a router to your Packet Tracer network and configure Inter-VLAN routing so that a PC in VLAN 10 can successfully ping a PC in VLAN

20.

Hint: Use router-on-a-stick configuration with sub-interfaces for each VLAN.

->Now add a router:

```
Router
|
trunk
|
+-----+
| Switch1|
+-----+
||
trunk
||
+-----+
| Switch2|
+-----+
```

Connect the router's **GigabitEthernet 0/0** to Switch1.

Configure the Switch1 port connected to the router as a trunk:

enable

configure terminal

interface gigabitEthernet 0/2

switchport mode trunk

switchport trunk allowed vlan 10,20,30

exit

Router-on-a-stick

On the router:

enable

configure terminal

```
interface gigabitEthernet 0/0
```

```
no shutdown
```

```
exit
```

VLAN 10

```
interface gigabitEthernet 0/0.10
```

```
encapsulation dot1Q 10
```

```
ip address 192.168.10.1 255.255.255.0
```

```
exit
```

VLAN 20

```
interface gigabitEthernet 0/0.20
```

```
encapsulation dot1Q 20
```

```
ip address 192.168.20.1 255.255.255.0
```

```
exit
```

VLAN 30

```
interface gigabitEthernet 0/0.30
```

```
encapsulation dot1Q 30
```

```
ip address 192.168.30.1 255.255.255.0
```

```
exit
```

Now configure the **default gateway** on the PCs:

VLAN	Gateway
VLAN 10	192.168.10.1
VLAN 20	192.168.20.1
VLAN 30	192.168.30.1

Test Inter-VLAN routing

From PC1:

```
ping 192.168.20.11
```

PC1 is VLAN 10 and PC2 is VLAN 20.

This should now work because the router performs **Inter-VLAN routing**.

Also test:

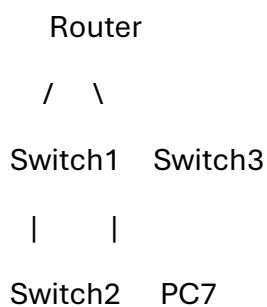
PC1 → PC3

PC2 → PC3

They should communicate successfully.

4. Set up static routes on the router so that a new network segment (e.g., a 'Creators' VLAN 40 on a third switch) can communicate with all existing VLANs, and test connectivity with ping.

->Add:



Create VLAN 40 on the switches:

enable

configure terminal

vlan 40

name Creators

exit

Assign PC7 to VLAN 40:

interface fa0/1

switchport mode access

switchport access vlan 40

exit

Give PC7:

IP Address: 192.168.40.11

Subnet Mask: 255.255.255.0

Gateway: 192.168.40.1

On the router, create the VLAN 40 sub-interface:

```
interface gigabitEthernet 0/0.40
```

```
encapsulation dot1Q 40
```

```
ip address 192.168.40.1 255.255.255.0
```

```
exit
```

Because all four VLAN networks are directly connected to the router, the router automatically knows:

192.168.10.0/24

192.168.20.0/24

192.168.30.0/24

192.168.40.0/24

So **static routes are not actually necessary when all VLANs terminate on the same router interface.**

If your assignment specifically requires a static route, you can demonstrate one using another router.

For example:

Router1

|

| 10.0.0.0/30

|

Router2

|

VLAN 40

On Router1:

```
ip route 192.168.40.0 255.255.255.0 10.0.0.2
```

On Router2:

```
ip route 192.168.10.0 255.255.255.0 10.0.0.1
```

```
ip route 192.168.20.0 255.255.255.0 10.0.0.1
```

```
ip route 192.168.30.0 255.255.255.0 10.0.0.1
```

Then test:

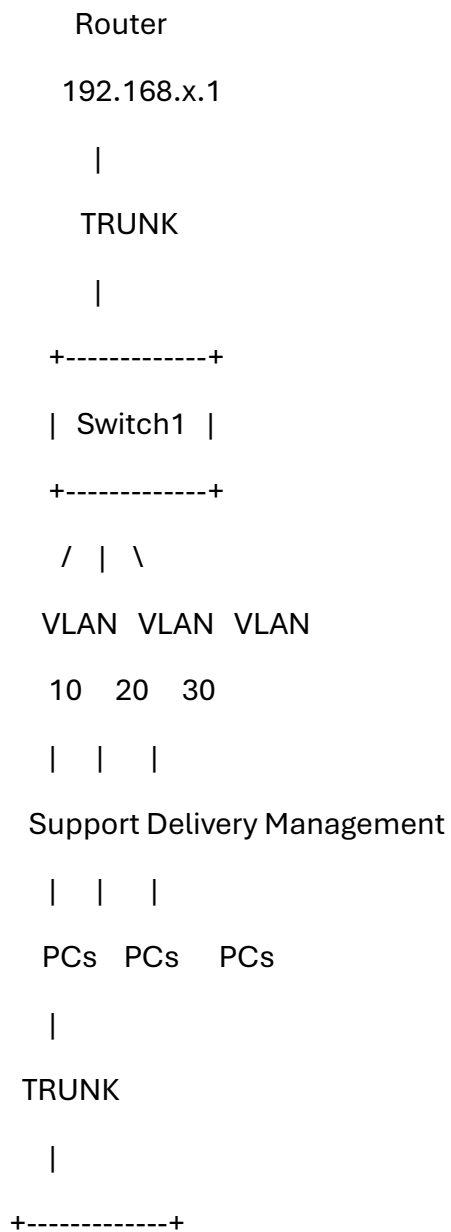
PC7 → PC1

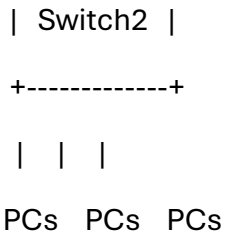
PC7 → PC2

PC7 → PC3

5. Design a mini network in Packet Tracer that mimics a Zomato-style office: create VLANs for 'Support', 'Delivery', and 'Management', configure trunking and static routing, and document your topology with a screenshot and a brief explanation of how devices in different VLANs communicate.

->You can make the final Packet Tracer project look like this:





VLAN plan

VLAN	Department	Network	Gateway
10	Support	192.168.10.0/24	192.168.10.1
20	Delivery	192.168.20.0/24	192.168.20.1
30	Management	192.168.30.0/24	192.168.30.1

Example PCs

Department	PC	IP
Support	PC1	192.168.10.11
Support	PC2	192.168.10.12
Delivery	PC3	192.168.20.11
Delivery	PC4	192.168.20.12
Management	PC5	192.168.30.11
Management	PC6	192.168.30.12

How communication works

Same VLAN:

Support PC1

|

Switch

|

Support PC2

The switch forwards the traffic within VLAN 10.

Different VLANs:

Support PC

|

VLAN 10

|

Switch

|

TRUNK

|

Router

|

VLAN 20

|

Switch

|

Delivery PC

The router receives the VLAN 10 traffic, routes it to the VLAN 20 network, and sends it back through the trunk to the destination PC.

Useful verification commands

On the switches:

show vlan brief

Check trunk:

show interfaces trunk

On the router:

show ip interface brief

Check routing table:

show ip route

Test connectivity:

ping 192.168.20.11

Expected final result

Test	Expected
Support PC → Support PC	✓ Success
Delivery PC → Delivery PC	✓ Success
Management PC → Management PC	✓ Success
Support → Delivery	✓ Success after Inter-VLAN routing
Support → Management	✓ Success after Inter-VLAN routing
Delivery → Management	✓ Success after Inter-VLAN routing
Creators → existing VLANs	✓ Success after routing

For your practical record, take screenshots of: **(1)** complete topology, **(2)** show vlan brief, **(3)** show interfaces trunk, **(4)** router show ip interface brief, and **(5)** successful ping results.